

Quantitative Trading and Price Impact

End-to-end pipeline with non-parametric impact
and multi-day carry backtest

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*Modular src/price_impact/ package | thin orchestration notebook | every backtest run writes to
saved/<name>/*

Pipeline at a glance

Seven assignment sections, eight modules, two enhancements

Modules vs assignment sections

Module	Responsibility	Section
data	top- N , σ /ADV	§2.1
impact_states	OU $\bar{I}$ (daily + multi)	§2.2/§2.3
fitting	OLS + NP + H -grid	§2.2
alpha	synthetic α	§2.4
strategy	OW/AFS/ext-OW	§2.5
backtest	Waelbroeck sim	§2.3
results	metrics + TCA + plots	§2.6
runner	BacktestConfig→disk	§2.6/§2.7

Design principles

- Single-responsibility modules; notebook is glue
- Same simulator runs **any** signed trade path
- Carry mode = one flag; downstream unchanged
- Every run dumps tables + plots to saved/<name>/

Two enhancements emphasised: **NP impact** with cross-stock γ shrinkage (§2.2) • **Multi-day carry** backtest engine (§2.3)

Solo submission – all sections by author.

Panel

- 2019 US equities, 10-s bins (monthly CSVs ~200 MB each)
- **Top-20 stocks** by full-year total $|q_t|$
- Final panel: ~11.0M bin rows, 250 trading days

Daily statistics (trailing 20-day rolling)

$$\sigma_{i,d}^2 = \overline{\text{Var}_t r_{i,d,t}}, \quad \text{ADV}_{i,d} = \overline{\sum_t |q_{i,d,t}|}$$

Used to normalise flow so impact coefficients are comparable across stocks and days.

In-sample / out-of-sample protocol

10 rolling windows:

- train on month m
- tune γ on $m+1$ (NP only)
- validate on $m+2$

for $m = 1, \dots, 10$.

```
build_panel(data_dir, year,
            top_n=20,
            lookback_days=20)
→ PanelData(bins, daily_stats,
            stocks)
```

Module data = 165 lines, deterministic, no I/O outside the bin CSVs.

Both models build the OU recursion

$$\bar{l}_{t+1} = (1 - \beta)\bar{l}_t + \tilde{q}_t, \quad \beta = \frac{\ln 2}{H_{\text{bins}}}.$$

They differ only in the **flow kernel** – how a child trade of size q_t pushes the state $\bar{l}$:

OW: $\tilde{q}_t = \sigma q_t / \text{ADV}$ (linear)

AFS: $\tilde{q}_t = \sigma \text{sgn}(q_t) \sqrt{|q_t| / \text{ADV}}$ (concave)

Why AFS is interesting. The model lets us *specify* the flow kernel $f(q)$ rather than inheriting the linear form by default. We pick the $\sqrt{q/V}$ because it lines up with the empirical $\sqrt{Q/V}$ meta-order law – mechanism explored on *slide 9* (Backtest engine), where the kernel choice becomes the operative modelling decision.

Optimisation problem. Joint

$$\max_{H, \alpha, \lambda} R_{\text{OOS}}^2(H, \alpha, \lambda).$$

For fixed H , OLS pins $(\hat{\alpha}, \hat{\lambda})$ in closed form, so the problem reduces to a **1-D scan over H** – the half-life that yields the best-fitting impact regression:

$$\max_H R_{\text{OOS}}^2(H, \hat{\alpha}(H), \hat{\lambda}(H)).$$

Regression target. At horizon $\tau = 6$ bins (60 s),

$$y_t = \alpha + \lambda x_t + \varepsilon_t,$$

with $x_t = \bar{l}_t - \bar{l}_{t-\tau}$, $y_t = (P_t - P_{t-\tau})/P_{t-\tau}$. Fit via additive sufficient statistics $(S_x, S_y, S_{xx}, S_{yy}, S_{xy}, n)$ – enables fast rolling re-fits and the H -grid sweep on the next slide.

Half-life grid search

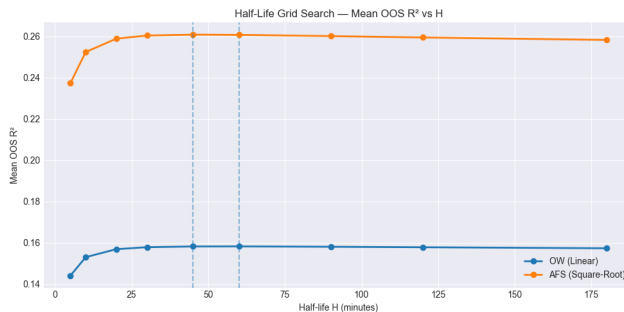
§2.2

H is a market property, not a model parameter

Mean OOS R^2 over
 $H \in \{5, 10, 20, 30, 45, 60, 90, 120, 180\}$
min:

H (min)	OW	AFS
10	0.1530	0.2525
30	0.1579	0.2604
45	0.1582	0.2609
60	0.1583	0.2608
90	0.1581	0.2602
120	0.1578	0.2595

OW peaks at $H=60$, AFS at $H=45$ – but not separate truths.



Why the midpoint? $H^* = 52.5$ min

- H is a **market property** – liquidity recovery rate. It should not depend on whether we model with OW or AFS flow
- Cross-model agreement is the test: both prefer 45–60 min within $< 10^{-4} R^2$. Midpoint is the natural single estimate of one universal market parameter

Non-parametric extension

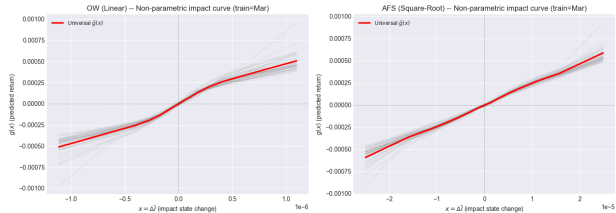
Method + estimated functional form

ENHANCEMENT #1

Replace the parametric $y = \alpha + \lambda x$ with a binned $g(x)$, applied to *both* OW and AFS (same procedure, different $x_t = \Delta \bar{I}$):

$$g_{i,b}^{\text{reg}}(\gamma) = \frac{S_{y,i,b} + \gamma \bar{g}_b}{n_{i,b} + \gamma}.$$

- $B = 15$ quantile bins on train month m (pooled across stocks)
- $\bar{g}_b$ = pooled universal curve across all stocks
- Per-stock $\hat{g}_{i,b}$ shrunk toward $\bar{g}_b$ with strength γ
- γ tuned on test month $m+1$, validated on $m+2$



Estimated $g(x)$ at train month $m=3$. Grey: per-stock $\hat{g}_i(\cdot)$; dashed: universal $\bar{g}(\cdot)$.

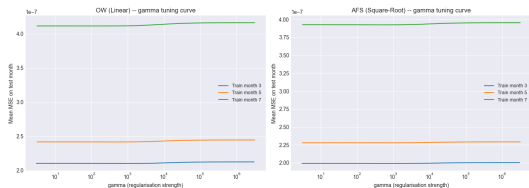
Left (**OW**, $x = \sigma \Delta \bar{I}_{lin}$): curves cluster around an approximately linear shape – but the model forces linearity, so the question is how much that constraint costs.

Right (**AFS**, $x = \sigma \Delta \bar{I}_{sqrt}$): curves are visibly **concave**, consistent with the empirical square-root impact regularity.

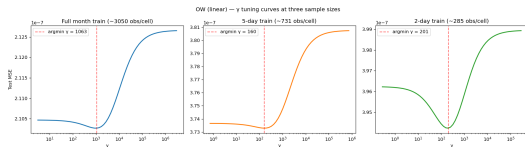
γ regularisation – regime-correctness

Flat in our data; kicks in for shorter training windows

Headline (1 month, ~ 3000 obs/cell)



Stress (train window 1 mo $\rightarrow$ 5 d $\rightarrow$ 2 d)



Test-month MSE vs γ , across multiple windows.
Curves **flat across 6 orders of magnitude** –
per-stock estimators are already precise; shrinkage
has nothing to fix.

Train window	Med. n	OW gain	γ^*/n
1 month	~ 3000	+0.0005	0.16
5 days	~ 720	+0.0025	0.56
2 days	~ 290	+0.0070	0.93

Estimator is **regime-correct**: trusts per-stock data when there's plenty, pools toward universal when starved. OOS gain grows $\sim 14\times$ as data shrinks $\sim 10\times$; shrinkage strength rises $6\times$. Headline flatness is a property of the data-rich regime, not a defect.

Headline OOS R^2 – results

§2.2

Apples-to-apples comparison of parametric vs non-parametric

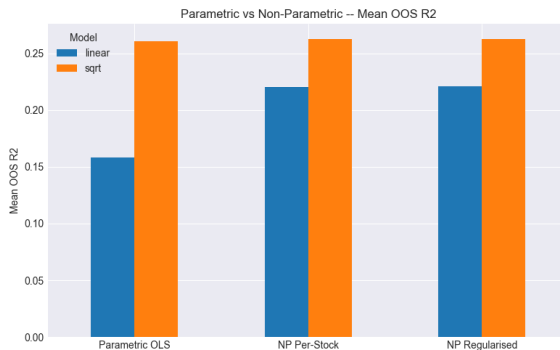
Pooled OOS R^2 (panel-wide null, same merged subset)

Model	Param.	NP Reg.	Δ
OW	0.1612	0.2206	+0.059
AFS	0.2689	0.2629	−0.006

Two findings

- **NP lifts OW by +5.9 pp** – linearity is a binding constraint
- **NP ties AFS** (−0.6 pp): the 1-parameter sqrt form sits on the efficient frontier; 15 parameters buy nothing

Aggregation matters: under per-stock R^2 averaging, NP appears to beat AFS by +0.2 pp – an artefact that disappears under pooled R^2



Mean OOS R^2 across rolling windows, by model and estimator. AFS dominates OW in every estimator family.

$$P_{\text{sim}}(t) = P_0(1 + \text{cumret}(t) + g_{\text{sim}}(t) - g_{\text{ref}}(t)), \quad g(t) = \lambda \bar{I}(t)$$

$\bar{I}^{\text{ref}}$ uses public-tape orderFlow; $\bar{I}^{\text{sim}}$ uses the engine's input trade path.

Modelling choice: why AFS sqrt? A constant-rate parent of size Q over T drives the OU recursion to equilibrium $\bar{I}^* = f(Q/T)/\beta$, so

$$\frac{\Delta P}{P_0} \propto f(Q/T).$$

The kernel's shape *propagates directly* to the parent (decay is linear – $1/\beta$ rescale, not reshape):

- **OW** ($f = q$): $\Delta P \propto Q/V$ – linear in size
- **AFS** ($f = \sqrt{q}$): $\Delta P \propto \sqrt{Q/V}$ – the empirical **Almgren / Bouchaud law**

Picking AFS at the engine level wires the empirical regularity in by construction.

Generic trades. Engine accepts any signed schedule via `trade_provider`:

`make_optimal_provider` (OW/AFS) and
`make_fixed_provider`
 (TWAP/VWAP/manual).

Carry modes.

- **daily** – reset $\bar{I}$ + inventory each session
- **multi** – persist with overnight decay $(1 - \beta)^{16^{\text{h}} \cdot 6}$ on $\bar{I}$; **inventory carries regardless**

Stock splits. Heuristic detector flags $|\Delta p|$, $|\Delta v|$ opposite-signed + stable notional. 2019 top-20: **none detected**.

Synthetic alpha (§2.4)

$$\alpha_t = r_t^h + y \frac{dW_t}{P_t}, \quad y = \sqrt{(\rho^{-2} - 1) \frac{\text{Var}(r^h)}{E[P^{-2}]h}}$$

Choosing $x=1$ in $\alpha = x r^h + y dW/P$ gives:

- **Unbiased:** $E[r^h \mid \alpha_t] = \alpha_t$
- **Target correlation:** $\text{Corr}(\alpha, r^h) = \rho$

We use $\rho = 0.05$, $h = 1$ bin (10 s); empirical ρ within $\pm 1\%$ of target.

OW closed form (§2.5)

$$X_t^* = \frac{\alpha_t \text{ADV}}{2\lambda\sigma}, \quad \kappa = \beta = \frac{\ln 2}{H_{\text{bins}}}$$

Position adjusts to X^* at speed κ ; ramped to zero over the final 30 min.

AFS variant. Same target rule per `project.ipynb`; sqrt nonlinearity enters via the simulator's $\tilde{q}$.

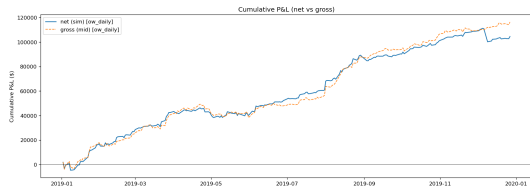
Ext-OW with $\lambda(t)$. **Placeholder.** Simulator already accepts a per-bin λ_t array; HJB-optimal trade rule is not κ -constant when λ varies – closed form pending.

Position trajectory sketch: X_t rises toward X_t^ at exponential speed κ , clips at $\pm 0.5\%$ ADV, ramps linearly to zero in the last 30 min. OW and AFS positions coincide; the cost accounting differs.*

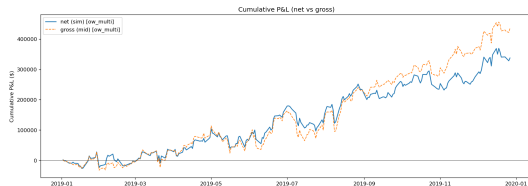
Performance – cumulative P&L

Daily-reset vs multi-day, OW model

§2.6



ow_daily: **Sharpe +4.30**, \$105k net, \$12k impact



ow_multi: Sharpe +1.61, **\$338k net**, \$98k impact

Net P&L (blue solid) marks to the Waelbroeck-distorted P_{sim} ; gross P&L (orange dashed) marks to the undistorted mid.

- **Daily-reset wins on Sharpe** – consistent recycling, tight drawdowns (−\$11k vs −\$83k)
- **Multi-day wins on dollars** – $\sim 3\times$ net P&L from accumulated inventory, but $\sim 8\times$ impact cost
- Turnover essentially identical between carry modes (\$84M) – only the *cost attribution* changes

TCA decomposition (totals)

§2.6

Net = Gross – Impact, with α rewards separated

Config	Gross P&L	Impact	Net P&L	Pred. α	Real. α	Turnover
ow_daily	116,311	11,787	104,524	44.6 M	116,311	84.0 M
ow_multi	435,399	97,813	337,586	44.6 M	435,399	84.0 M
afs_daily	119,205	56,417	62,787	44.6 M	119,205	86.6 M
afs_multi	482,557	129,999	352,558	44.6 M	482,557	86.6 M

- **Impact cost = Gross – Net** – well-defined under Waelbroeck construction
- **Realised α reward** $\sum_t \text{fwd_ret}_t \cdot X_t \cdot P_t \equiv$ Gross P&L by construction
- **Predicted α reward** $\sum_t \alpha_t \cdot q_t \cdot P_t$ – dominated by α 's noise component at $\rho = 0.05$ (only $\rho^2 = 0.25\%$ of $\text{Var}(\alpha)$ is signal); informative *across* configs, not for level
- Slippage components needing fill-price detail (arrival vs VWAP) intentionally omitted

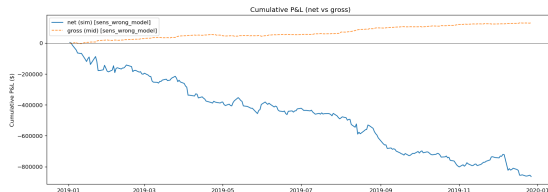
Live sensitivity sweeps (OW, daily-reset)

Scenario	Sharpe	Net P&L
Baseline	+4.30	\$105k
Wrong model (OW $\rightarrow$ AFS)	-3.48	-\$863k
Signal delayed 1 min	-0.32	-\$8k
$\rho = 0.01$	+0.45	\$11k
$\rho = 0.10$	+9.22	\$251k
$H = 30$ min	+5.91	\$205k
$H = 120$ min	+3.26	\$53k

Takeaways

- **Wrong model catastrophic** – -\$863k
- **1-min delay flips the sign** – α is short-horizon
- ρ scales Sharpe linearly:
 $\{0.01, 0.05, 0.10\} \rightarrow \{0.45, 4.30, 9.22\}$
- H is forgiving: $\pm 2\times$ stays in $[+3.3, +5.9]$

Anatomy of the worst case. Fit OW $\hat{\lambda}$ on the panel, then trade as if impact were AFS (sqrt). Year-long net P&L:



Gross alpha-capture (orange dashed) still builds steadily – the signal is real. But OW-sized positions under AFS impact pay ~\$995k in impact cost, dwarfing gross P&L. Net (blue) collapses early.

$\Rightarrow$ **Model risk dominates signal risk.**

What we learned

Re-stating the two enhancements with one bullet each

Enhancement 1 – Non-parametric impact with γ regularisation (§2.2)

Apples-to-apples pooled OOS R^2 : **OW** 0.161 $\rightarrow$ 0.221 (+5.9 pp, linearity binding); **AFS unchanged at 0.263** (square-root on the efficient frontier; NP buys nothing). γ shrinkage immaterial in the data-rich headline regime; γ^*/n rises 6 $\times$ and **OOS gain 14 $\times$** as the training window shrinks 10 $\times$ – the estimator behaves correctly under stress.

Enhancement 2 – Multi-day carry backtest (§2.3)

Carrying $\bar{I}$ and inventory across sessions makes the cost of impact *compound*: in the OW config, **impact cost \$12k $\rightarrow$ \$98k** ($\sim 8\times$), gross P&L **3 $\times$ larger**. Daily-reset wins on Sharpe (+4.30 vs +1.61); multi-day wins on absolute P&L. Engine handles both via a single flag; downstream code, TCA columns, and plotting are unchanged.

Honest limitations: overnight α (§2.4 to-expand) and extended-OW $\lambda(t)$ closed form (§2.5 to-expand) are flagged but not implemented; the simulator side of $\lambda(t)$ is already plumbed via `ImpactModel.lam_t_lookup`.

Thank you

Questions?

Code: `src/price_impact/` | Notebook: `final_pipeline.ipynb` | Runs: `saved/<name>/`